### Self-Check

7. Which is longer for a reliable device?
   a. MTBF
   b. MTTR
   c. SPoF
   d. SSOT
8. Which change is designed to close a security vulnerability without affecting other components of an application?
   a. Patch
   b. Rollback
   c. Upgrade
   d. Installation
9. Which change management principle must you apply to all changes regardless of scope or impact?
   a. Acquire review board approval.
   b. Apply the change as soon as possible.
   c. Submit a change request document.
   d. Document the change.

Check your answers at the end of this chapter.

# Summary

## Section 2-1: Components of Structured Cabling

- The ANSI/TIA-568 structured cabling standards describe uniform, enterprise-wide cabling systems regardless of who manufactures or sells the various parts used in the system.
- For most situations, the device that marks where an ISP's network ends and the organization's network begins is the demarc, or demarcation point.
- Connections branching out from the MDF include Ethernet cables connecting to nearby work areas, large cables running to IDFs in other buildings or on other floors of the same building, and the incoming connection from the service provider's facility.
- VoIP (Voice over IP), also known as IP telephony, is the use of any network (either public or private) to carry voice signals using TCP/IP protocols. A common application layer protocol used by voice gateways to initiate and maintain connections is the signaling protocol SIP (Session Initiation Protocol).
- Cable management best practices include minimizing exposed wires at each termination, ensuring continuity, loosely cinching cables, covering cables, organizing cables with cable trays and conduits, reducing cable exposure to EMI, and keeping good documentation.
- HVAC sensors are often connected as networked devices themselves. HVAC sensors and the software to control them are an example of an ICS (industrial control system), which acquires real-time data from a physical system and manages the system or presents the data to humans to monitor and manage the system. A more complex type of ICS is SCADA (supervisory control and data acquisition), which is the interface for a widespread control system. OT (operational technology) is hardware and software that interacts directly with physical infrastructure and devices through monitoring and control of these systems.

## Section 2-2: Network Documentation

- Network diagrams are graphical representations of a network's devices and connections. These diagrams might show logical topology, IP address reserves, names of major network devices, types of transmission media, and physical layout such as a floor plan or rack diagram.
- A network diagram is often the product of network mapping, which is the process of discovering and identifying the devices on a network. One of the simpler and most popular tools is Nmap (Network Mapper).
- A rack diagram shows the devices stacked in a rack system and is typically drawn to scale. Rack diagrams are helpful

when planning a rack installation or when troubleshooting equipment installed in a rack.

- Essential network documentation covers logical and physical connections on a network; inventory management; IP address utilization; vendors (such as contact information, warranty information, service agreements, and troubleshooting instructions); and internal operating procedures, policies, and standards. For example, many corporations establish SOPs (standard operating procedures) to ensure consistency as different employees perform the same complex tasks.
- A baseline configuration of network devices and their software is a change management concept that defines a stable state to be achieved and maintained before attempting future changes.
- Systematically name and label network devices and connections. A good naming convention can serve double duty by including essential information about each device while also allowing technicians to identify and track each device more easily.
- Important business documents in network administration include the MOU and the SLA. An MOU (memorandum of understanding) documents the intentions of two or more parties to enter into a binding agreement, or contract, and is sometimes used between an informal handshake and the legally binding signatures on contracts. An SLA (service-level agreement) is a legally binding contract or part of a contract that defines, in plain language and in measurable terms, the aspects of a service provided to a customer.

## Section 2-3: Asset Management

- Devices on a network typically have a calculated MTBF (mean time between failures). This is the average amount of time that will pass for devices exactly like this one before the next failure is expected to occur. Once a device fails, there is an average amount of time required to repair the device. This is called MTTR (mean time to repair), and this cost must also be considered.
- Redundancy refers to an implementation in which more than one component is installed and ready to use for storing, processing, or transporting data. Redundancy is intended to eliminate any SPoF (single point of failure).
- The process of designing, implementing, and maintaining these systems is called the system life cycle. All these systems rely on some type of configuration, which consists of various values defined to determine how the system operates.
- Four common types of software changes include installations, patches, software upgrades, and rollbacks after a failed change. Hardware changes might include adding new equipment or replacing old equipment. Hardware changes might also include firmware updates. Balance the risks with the potential benefits, especially if the update includes security enhancements that close critical vulnerabilities.
- Required change management processes and related documentation are designed to protect the person making the change, users, managers, and the organization so that changes don't unnecessarily disrupt normal work flow or put undue responsibility for a problem on any one person.

# Key Terms

For definitions of key terms, check the Glossary.

automatic failover
automation
baseline configuration
cable management
CM (configuration management)
configuration
configuration drift
crosstalk
decommissioning
demarc (demarcation point)
EF (entrance facility)
EMI (electromagnetic interference)
EOL (end-of-life)
EOS (end-of-support)
fiber-optic cable
golden configuration
hot-swappable
ICS (industrial control system)
IDF (intermediate distribution frame)
inventory management
KVM (keyboard, video, and mouse) switch
licensing restrictions
life cycle management
maintenance window
MDF (main distribution frame)
MOU (memorandum of understanding)
MSA (master service agreement)
MTBF (mean time between failures)
MTTR (mean time to repair)
network diagram
Nmap (Network Mapper)
OT (operational technology)
patch
patch management

patch panel
plenum
port-side exhaust switch
port-side intake switch
project
project management
rack diagram
redundancy
RFP (request for proposal)
rollback
SCADA (supervisory control and data acquisition)
SIP (Session Initiation Protocol)
SLA (service-level agreement)
SOP (standard operating procedure)
SOW (statement of work)
SPoF (single point of failure)
SSOT (single source of truth)
STP (shielded twisted pair)
structured cabling
system life cycle
upgrade
UTP (unshielded twisted pair)
version control
VoIP (Voice over IP)
wiki

# Acronyms Checklist

The acronyms in Table 2-5 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

**Table 2-5** Chapter 2 acronyms

| Acronym | Spelled out |
|---|---|
| EOL | end-of-life |
| EOS | end-of-support |
| EULA | end-user license agreement |
| ICS | industrial control system |
| IDF | intermediate distribution frame |
| MDF | main distribution frame |
| MTBF | mean time between failure |
| MTTR | mean time to repair |

| Acronym | Spelled out |
|---|---|
| OT | operational technology |
| SCADA | supervisory control and data acquisition |
| SIP | Session Initiation Protocol |
| SLA | service-level agreement |
| STP | shielded twisted pair |
| USB | universal serial bus |
| UTP | unshielded twisted pair |
| VoIP | Voice over IP (Internet Protocol) |

# Review Questions

1. A technician from your ISP has arrived to help you troubleshoot a weak WAN connection. To what location do you take them?
   a. IDF
   b. Work area
   c. CEO's office
   d. EF
2. Which of the following devices are you likely to find in the MDF? Choose all that apply.
   a. Routers
   b. Switches
   c. Network printer
   d. KVM switch
3. Which device converts signals from a campus's analog phone equipment into IP data that can travel over the Internet?
   a. Legacy PBX
   b. VoIP phone
   c. Voice gateway
   d. Dedicated telephone switch
4. What command invokes Nmap at a command line?
   a. `zenmap`
   b. `nmap`
   c. `ping`
   d. `cli`

5. What information should never be included in the name of a web server?
   a. Building location
   b. Purpose on the network
   c. Username
   d. Name of department
6. Which document should be adjusted if your budget for a project has changed?
   a. SLA
   b. RFP
   c. MOU
   d. SOW
7. A network administrator stores passwords in a safe that no one else has access to. This is a(n) _____.
   a. redundancy
   b. password manager
   c. SOP
   d. SPoF
8. Configuration monitoring is intended to prevent _____.
   a. a baseline configuration
   b. a golden configuration
   c. configuration drift
   d. configuration versions
9. Historic configurations are typically kept for the purpose of _____.
   a. auditing
   b. automation
   c. discovery
   d. compliance
10. In what period should you schedule a network change?
   a. Maintenance window
   b. Rollback
   c. Automatic failover
   d. Decommissioning window

# Scenario-Based Questions

## Question 2-1

A transceiver was recently damaged by a lightning strike during a storm. How might you decide whether the ISP is responsible for replacing this device, or whether your company must foot the bill and pay for repairs?

a. Determine whether the device is located on the ISP's side of the demarc.
b. Examine the manufacturer information on the device's label.
c. Examine purchase records for the device to determine when it was acquired.
d. Determine what kinds of cables are connected to this device.

## Question 2-2

You need to connect a new network printer to a nearby wall jack. The printer will be used for printing documents, spreadsheets, and reports for a small project team in nearby offices. What kind of cable should you use?

a. Fiber-optic cable
b. Patch cable
c. Backbone cable
d. Plenum-rated cable

## Question 2-3

After adding three new servers and a network printer to your network, you're having problems identifying which IP address belongs to each device. You've decided to run an Nmap scan of your network. Which TWO apps could you use to perform this task?

a. Zenmap
b. Microsoft Edge
c. Microsoft Visio
d. PowerShell

## Question 2-4

You've been assigned the task of inventorying all IT assets in your department. You've been given a stack of purchase receipts, some user manuals, and an old spreadsheet of items scheduled for maintenance. What is the first step you should take to begin developing updated and relevant documentation for ongoing inventory management?

**a.** Interview users.
**b.** Identify network requirements.
**c.** List an administrative account's username and password for each device on the network.
**d.** List all components on the network.

## Question 2-5

You're shopping for a new router, and you've found three models that will meet your needs and your budget. After some recent problems with another router on your network, which of the following metrics do you want to maximize to avoid similar problems with the new router?

**a.** EOS
**b.** MTBF
**c.** EOL
**d.** MTTR

# Hands-On Projects

**Note 2-12** | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

**Note to Instructors and Students:** A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

## Project 2-1: Create a Wiki

**Estimated time:** 30 minutes (+ up to an hour to wait on Wikidot activation email)
**Objective 3.1:** Explain the purpose of organizational processes and procedures.
**Group work:** This project includes enhancements when assigned as a group project.
**Resources:**
- Internet access

**Description:**
You can collect information from members of your team using a wiki creator to build your own wiki. There are many good wiki tools that require a purchase or subscription, such as Google Sites and SharePoint. These paid apps provide impressive features that really shine when you're collaborating with several people or making your wiki public as part of your business front. A free app such as Wikidot (wikidot.com), though, can give you all the features you need for tracking your own project information in this course. This experience will also help you better understand which features of a wiki are most important to you.

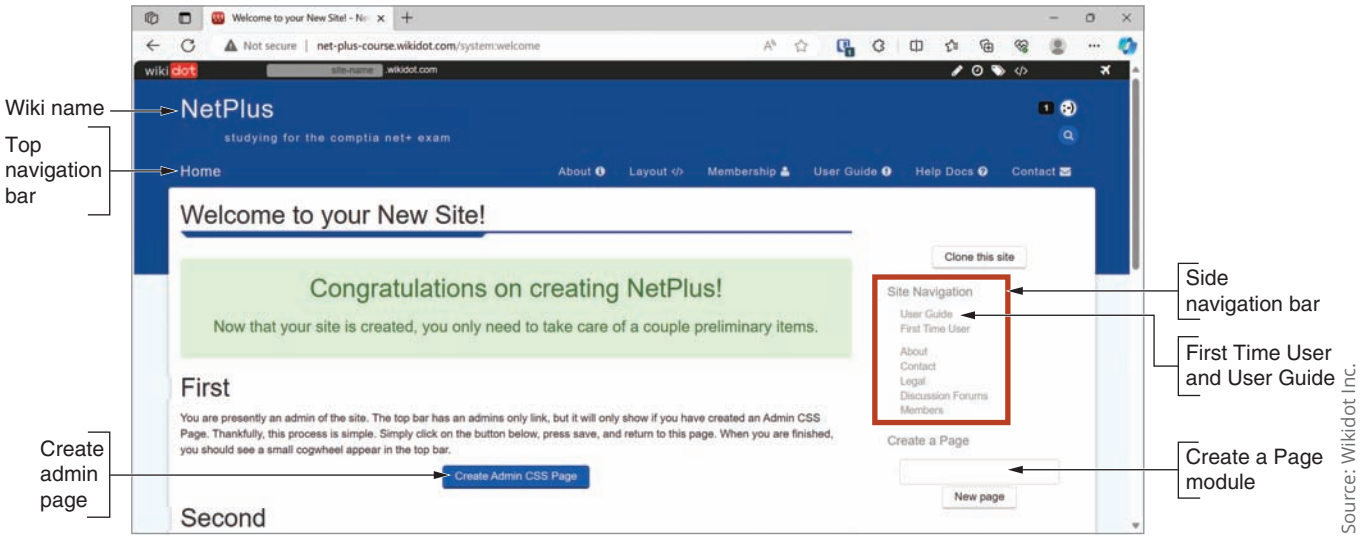
Complete the following steps to create your own wiki:

1. Go to **wikidot.com** and create an account. **Tip:** Do not use an email address with numbers in it, as the Wikidot system is likely to reject this account with no warning. Additionally, many users have experienced problems creating a Wikidot account using the Facebook authentication option.
2. Store your account information in your Bitwarden vault. Check your email to activate your Wikidot account. You should receive an activation email within about an hour. If it takes longer, you might need to try again, or use a different email address.
3. After activating your Wikidot account, click the **Sites** tab. In the left pane, scroll down and click + **Create site**.
4. Give your wiki a title and web address, and make sure the **Standard Template** is selected.

**Note 2-13** | You can choose any template for a later wiki, or you can later change the template for this wiki after completing this project.

5. Select an access policy. Unless directed otherwise by your instructor, choose the **Private** option.
6. Confirm that you have read and agreed to the Terms of Service, and then click **Get my Wikidot site**. Your Wikidot site is created with generic information that will help you get started (Figure 2-35). What title and web address did you give your wiki?

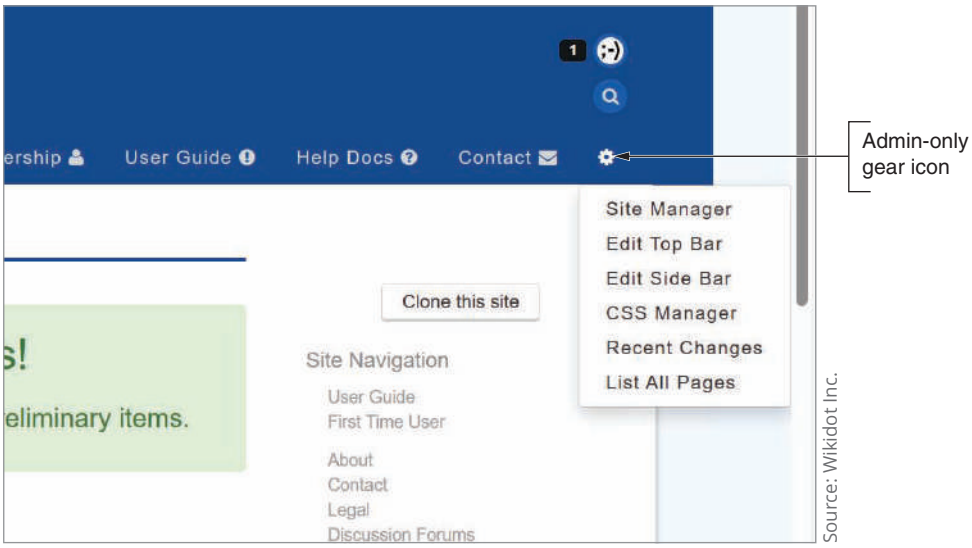
**Figure 2-35** Use a wiki to collect project or network information that all your team members can access




Source: Wikidot Inc.

7. To see the admin link in the top navigation bar, click **Create Admin CSS Page** under the "First" section on the welcome page. Without making any edits to the page, click **Save** at the bottom of the page. There should now be a gear icon in the top navigation bar (Figure 2-36). Click it to see the admin navigation menu. You'll come back to this menu soon.

**Figure 2-36** The gear icon accesses the admin navigation menu




Source: Wikidot Inc.

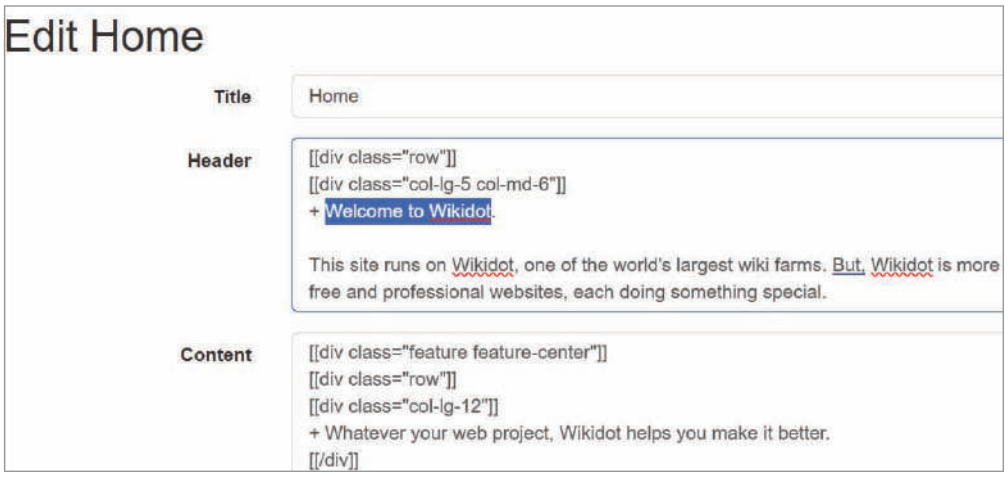
8. In the side navigation bar, click **First Time User** and read the information on this page. According to the documentation, what feature in the Standard Template makes page creation simple?
9. Click **User Guide** in the side navigation bar and skim through this page to understand how the Site Manager works and how to edit pages. What does the Site Manager let you do?
10. Click each tab along the top navigation bar and skim the contents: **Home**, **About**, **Layout**, **Membership**, **User Guide**, **Help Docs**, and **Contact**. This will help you become familiar with the pages included in this template. Each of these pages belongs to your personal wiki site and includes default information you can change later.
11. Click the **Home** link to go back to the Home page. At the bottom of the page, click **Edit** (Figure 2-37).

**Figure 2-37** Page tools at the bottom of each wiki page

Source: Wikidot Inc.

12. In the Header box, change the text *Welcome to Wikidot* (Figure 2-38) to say **This is my Wikidot.** Save your changes. **Take a screenshot** of your new Home page; submit this visual with your responses to this project's questions.

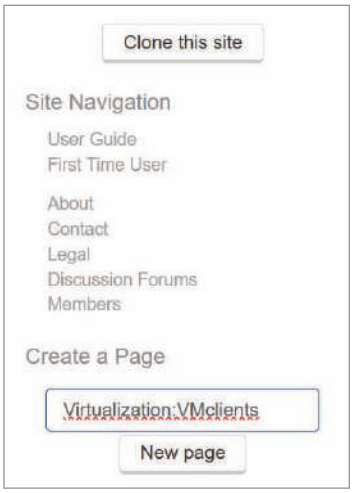
**Figure 2-38** Make changes to the text on any page




Source: Wikidot Inc.

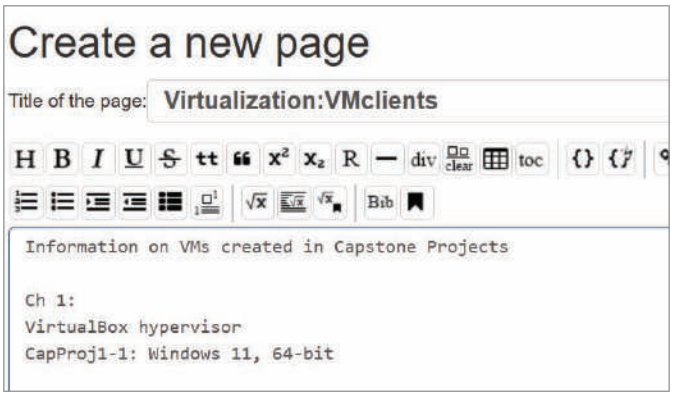
13. Click the **About** link to go to the About page. In the side navigation bar, in the "Create a Page" module, type **Virtualization:VMclients** (Figure 2-39). Notice there is no space before or after the colon. This will create a new page named "VMclients" in the category "Virtualization". Click **New page**.
14. You have 15 minutes to complete changes to this page. Type the information for the Capstone Project you completed in Chapter 1 (Figure 2-40)—substitute the information for your own VM, such as the name you gave the VM and the OS you installed on it. Click **Save**.

**Figure 2-39** Create a new category and page in a wiki




Source: Wikidot Inc.

**Figure 2-40** Add information for your Chapter 1 VM




Source: Wikidot Inc.

15. To see a list of all pages in your wiki, click the gear icon in the top bar, and then click **List All Pages**. You will continue to add pages and content in later chapters. For now, click your new **Virtualization:VMclients** page in the list. **Take a screenshot**; submit this visual with your responses to this project's questions.
16. Click the gear icon and click **Site Manager** to go to the Dashboard.

17. In the left pane, click **Security** and then click **Access policy**. If necessary, scroll up to see your current access policy. What access mode is selected? Scroll down and notice in the blue box that you can apply for a free educational upgrade. This is optional and is not required for this course. Ask your instructor whether you should pursue the upgrade.
18. **For group assignments:** Invite your group members to your wiki. To do this, in the left pane, click **Back**, click **Members**, and then click **Invitations**. On the Invite members tab, enter the username of each group member to invite them (the search might take a few minutes). Accept your group members' invitations. To do this, go to the **Messages** tab of your account and click **Invitations**. Click the invitation (not the inviter's profile name) and click **Accept**. After each group member accepts your invitation, in the left pane, click **Members List**. View your page's members, explore the levels of permissions you can manage for each member, and send each member a private message.
19. Click through the other settings and options, and make changes as desired. When you're finished, click your wiki's name in the upper-left corner next to the Wikidot logo to return to your wiki.

## Project 2-2: Install and Use Wireshark

**Estimated time:** 45 minutes
**Objective 3.1:** Explain the purpose of organizational processes and procedures.
**Resources:**
- A computer or user account with application installation rights
- Internet access
- Access to a private network either owned by the student or which the student has explicit permission to scan

**Description:**
Wireshark is a free, open-source network protocol analyzer that can help demystify network messages for you and help make the OSI model easier to understand. For some students, using Wireshark for the first time can be an epiphany experience. It allows you to study the OSI layers, much of the information that is added to every message, and so many of the messages that go back and forth just to bring up a webpage or simply to connect to the network. It all becomes much more real and concrete when you see how many messages Wireshark collects during even a short capture.

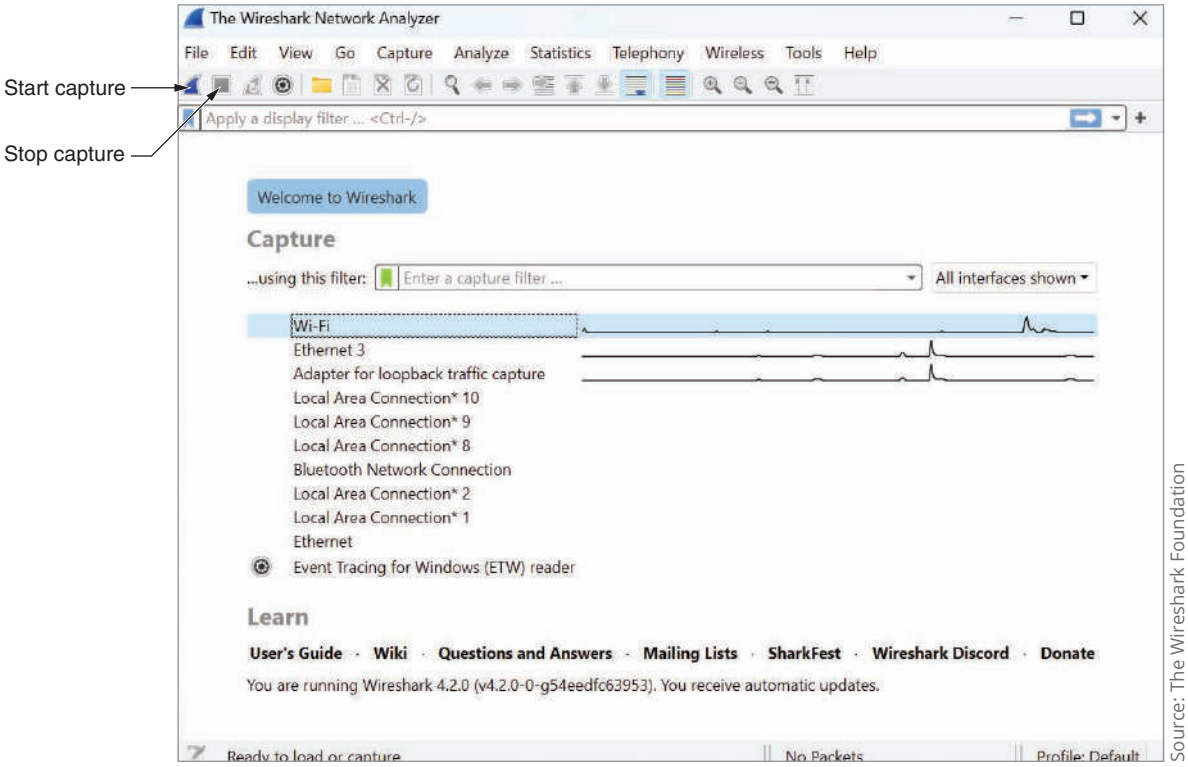
**Caution** | Take note that it is illegal to scan a network you don't own or don't have permission to scan with Wireshark. Do not use Wireshark on public Wi-Fi networks at all. Also don't use Wireshark on any network you don't own unless you have written permission from the owner to capture and analyze network communications using Wireshark. Make sure that everyone else using the network is aware you will be performing Wireshark captures and when, and the kind of information you will be able to access from those captures.

In this project, you will install Wireshark and learn the basics of how it works. Complete the following steps:

1. Open a browser and go to **wireshark.org**. Download and install the current stable release, using the appropriate version for your OS. In the Wireshark setup window, accept all default settings; for example, you do *not* need USBPcap. If needed, reboot your computer to complete the Wireshark installation.

2. When installation is complete, open **Wireshark**.
3. In the Wireshark Network Analyzer window, select your network interface from the list. Then click the shark-fin icon to start the capture (Figure 2-41).

**Figure 2-41** The Wireshark Network Analyzer window



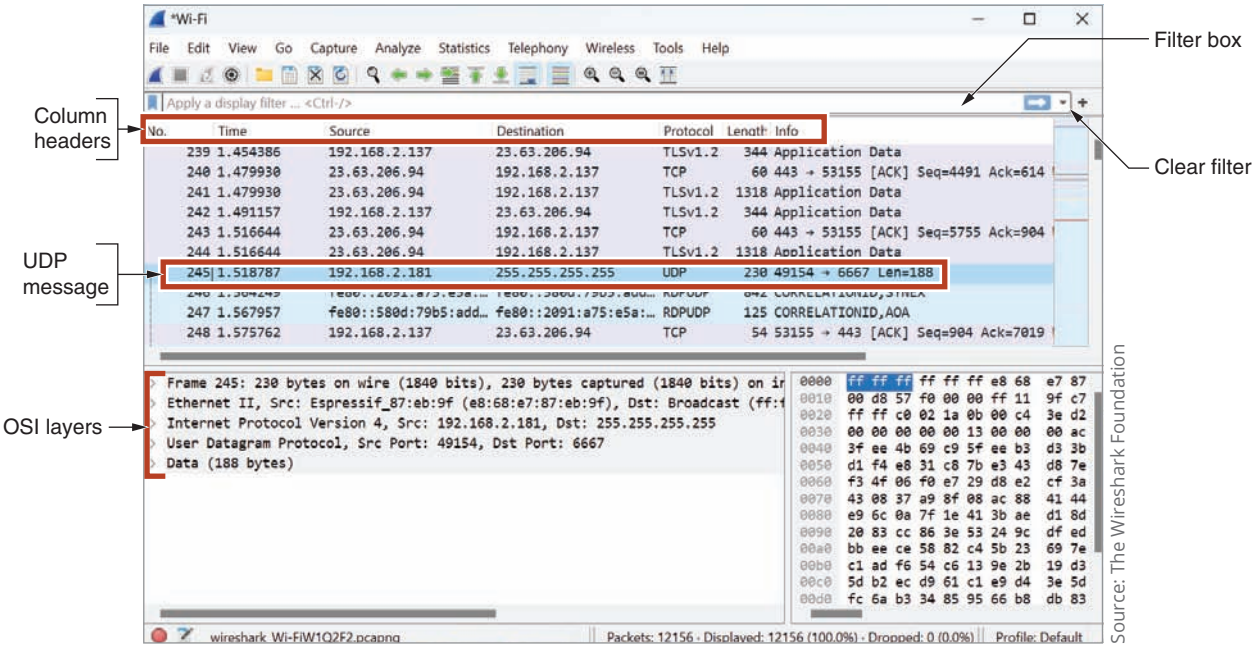

Source: The Wireshark Foundation

4. While the capture is running, open your browser and go to **cengage.com**. Then open a PowerShell or Command Prompt window, type `ping 8.8.8.8`, and press **Enter**. After the ping completes, on the Wireshark capture window, click the red box on the command ribbon to stop the capture.

Scroll through some of the messages you've captured. You can adjust the pane sizes by grabbing a border between them and dragging. Expand the top pane so you can see more of the captured messages at one time. Let's start to decode this blur of numbers and letters.

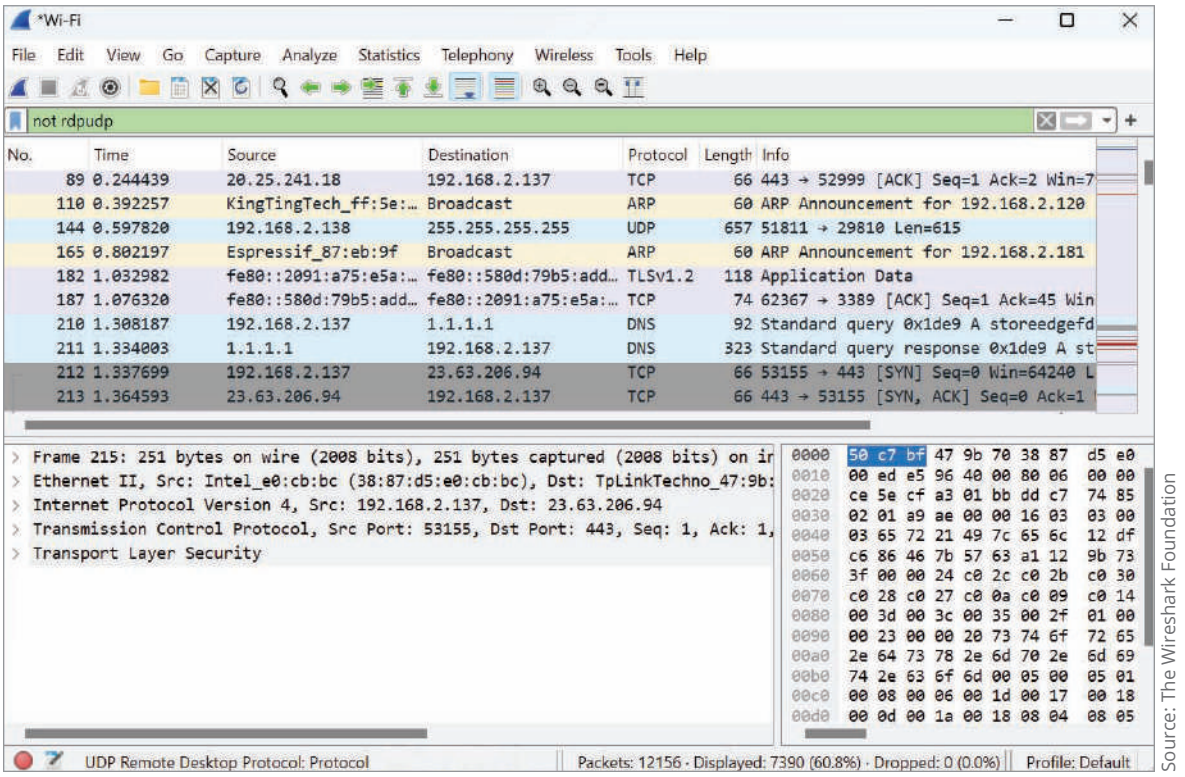
5. Notice the column headers along the top of the capture (Figure 2-42). Of particular interest are the Source and Destination columns, the Protocol column, and the Info column. Find a UDP (User Datagram Protocol) message that has an IPv4 Source address and click on it. **Take a screenshot** of your capture with the UDP message selected; submit this visual with your responses to this project's questions.
6. In the middle pane, click each line to expand that layer's information. What pieces of information stand out to you? Which device on your network do you think sent this message, and which device(s) received it?

**Figure 2-42** A Wireshark capture




Color highlighting can make it easier to spot different protocols. Notice in Figure 2-43 that TCP (Transmission Control Protocol) messages are a light lavender, ARP (Address Resolution Protocol) messages are a pale yellow, and UDP and DNS (Domain Name Services) messages are a light blue. The protocol names are listed in the Protocol column. Note that if you have trouble distinguishing colors, you can choose colors or shades that work for you. For example, you might choose a very dark shade that stands out against

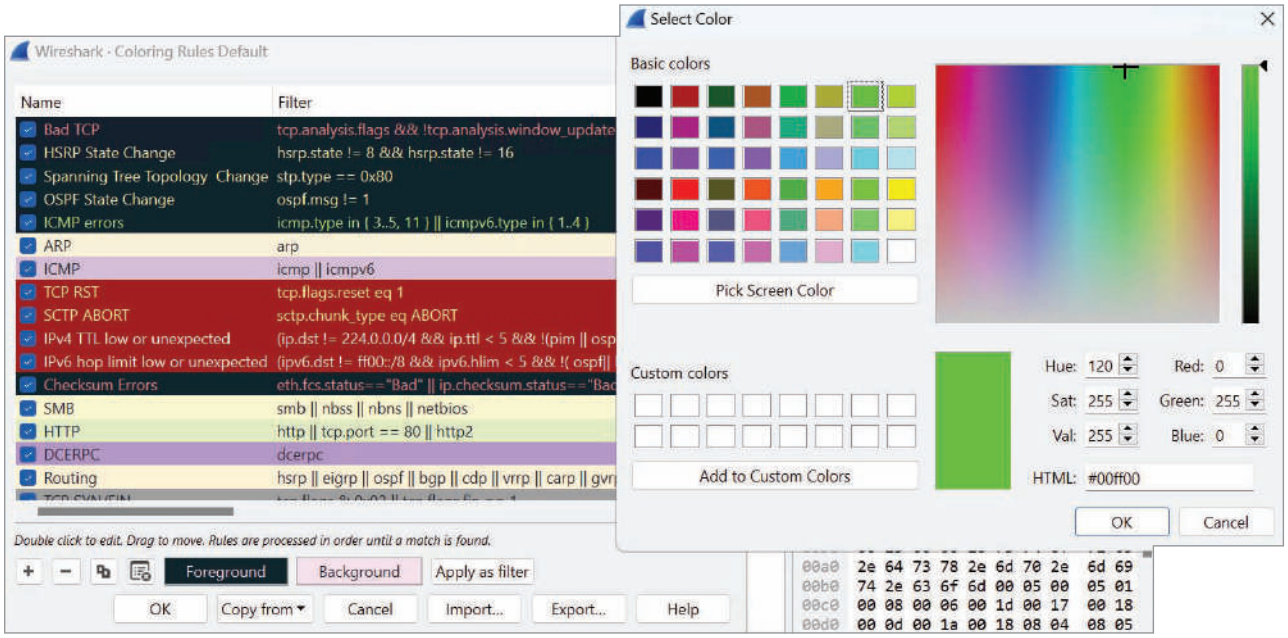
**Figure 2-43** Different highlight colors correspond to different protocols



lighter shades. If necessary, you could also use a phone app to help, such as Be My Eyes (bemyeyes.com) that provides AI-powered visual assistance or pairs users with a normally sighted volunteer to help identify colors, Color Blind Pal (colorblindpal.com) that provides descriptive information and offers a filter tool, or the Pixolor app that identifies colors of pixels in an image (website developers often use apps like this to define branding colors).

7. To display a list of currently assigned highlight colors and to adjust these assignments, on the main toolbar, click **View** and then click **Coloring Rules**. Here, you can change the priority for matching protocols within a message to colors in the output pane (because more than one protocol is used in each message), and you can assign colors that are easier to spot. In Figure 2-44, the background color for ICMP is changed to a bright green. When you're happy with your color selections, click **OK**.

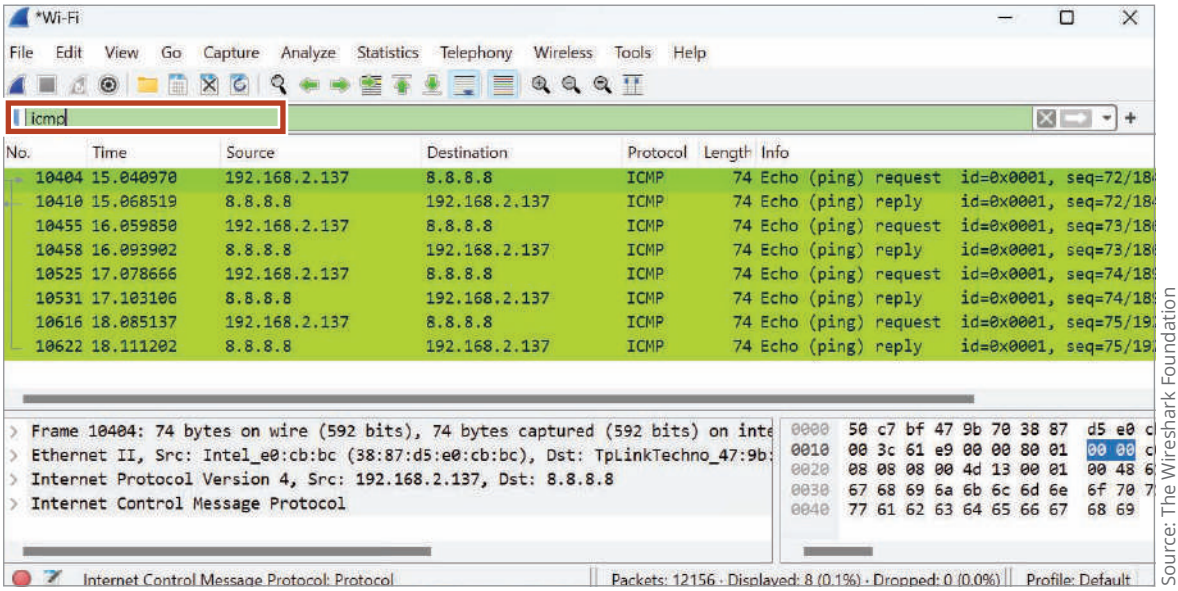
**Figure 2-44** Choose colors that are easier for you to work with




Source: The Wireshark Foundation

8. To filter for a particular kind of message in your capture, type the name of the protocol in the Filter box (identified in Figure 2-44). Figure 2-45 shows a filter for ICMP messages, which are currently highlighted in bright green. These ICMP messages were generated when pinging another host on the network. Try filtering for other protocols you've read about in this course already, such as HTTP, and see how many different types you can find in your capture. Click **X** to clear filters between searches. Which protocols did you find?
9. To compare which OSI layers are represented by each of these protocols, apply a slightly more complicated filter where you can see both DNS messages and ICMP messages in the same search. Enter the following phrase into the Filter box: **dns or icmp**.

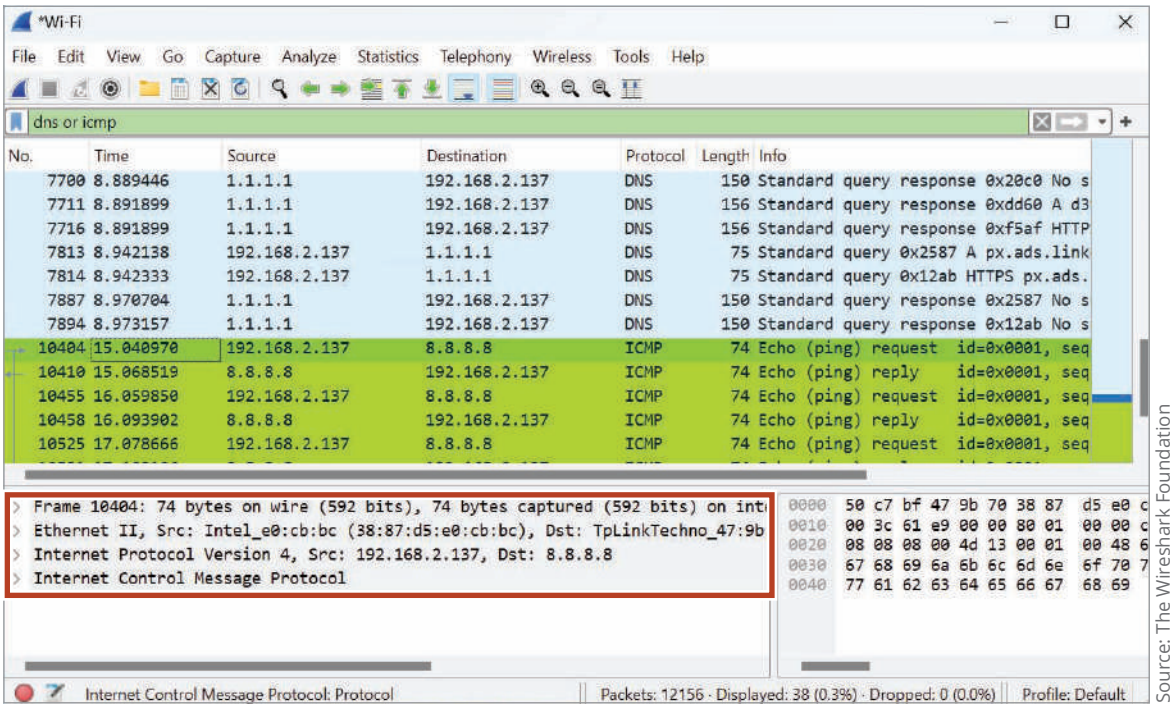
**Figure 2-45** Use filters to narrow your search




Source: The Wireshark Foundation

10. Click an ICMP message and count the layers of information available in the middle pane. In Figure 2-46, there are four layers of information, which correspond to layer 2 (Frame and Ethernet II) and layer 3 (Internet Protocol Version 4 and Internet Control Message Protocol).

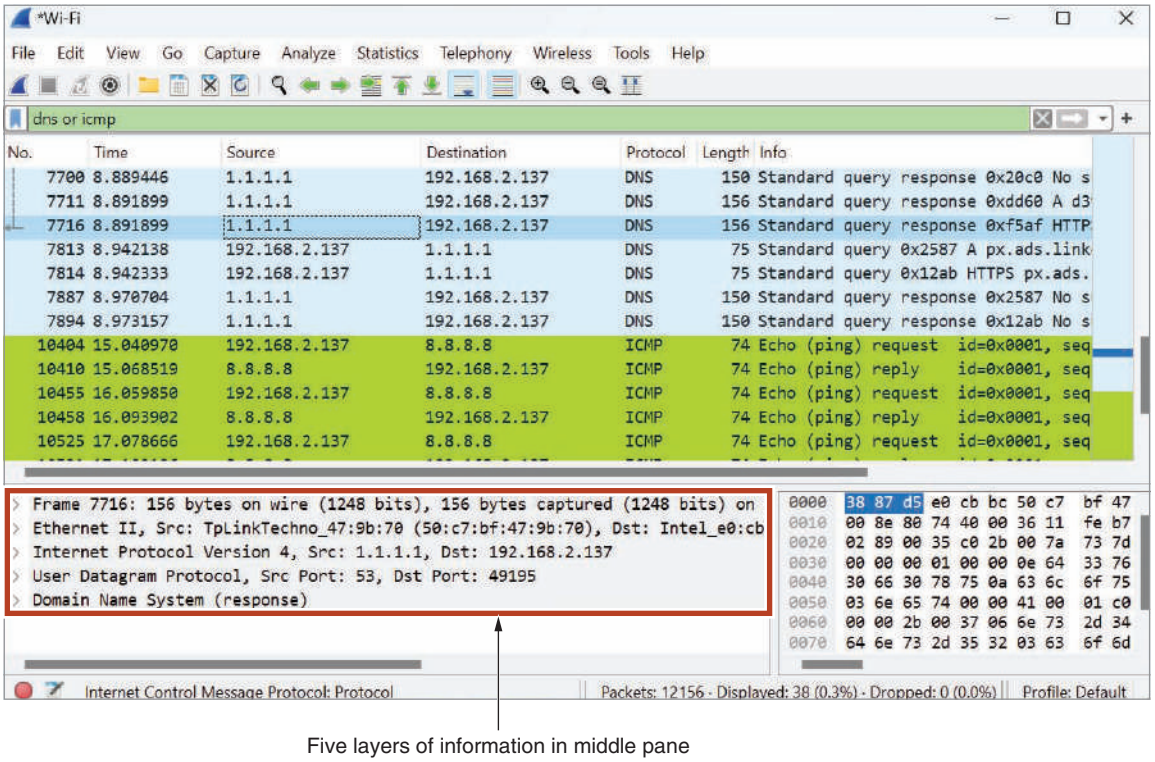
**Figure 2-46** Use the middle pane to dig into each layer's headers




Source: The Wireshark Foundation

11. Examine a DNS message. Figure 2-47 shows five layers of information in the middle pane. This time, layer 7 (Domain Name System) and layer 4 (User Datagram Protocol) are represented, in addition to layer 3 (Internet Protocol Version 4) and layer 2 (Ethernet II and Frame).

**Figure 2-47** This DNS message uses UDP at the transport layer and IP at the network layer




Source: The Wireshark Foundation

Five layers of information in middle pane

12. In your wiki, add a new page titled **Applications:Wireshark**. Indicate the chapter and project number for this installation, the computer you used for this project, a brief description of what you learned, and any other information you might find helpful when using Wireshark later.

# Capstone Projects

**Note 2-14** | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

**Note to Instructors and Students:** A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

## Capstone Project 2-1: Set Up an Ubuntu Desktop VM

**Estimated time:** 45 minutes

**Objective 5.5:** Given a scenario, use the appropriate tool or protocol to solve networking issues.

**Resources:**

- Access to the same computer used to complete Capstone Project 1-1
- Internet access
- If desired, instructor can provide Ubuntu Desktop image file

**Description:**

In the Chapter 1 Capstone Project, you created a virtual machine using Oracle VirtualBox. In this Capstone Project, you create a second VM in your virtual network and install Ubuntu Desktop on the VM. In Chapter 3, you'll install Ubuntu Server in your network. Ubuntu is a well-known version of Linux and offers both desktop and server editions. To effectively troubleshoot network problems, you will need to be able to use tools specific to Linux, and these VMs will give you the opportunity to practice working with Linux tools. For these VM projects, you can use your choice of hypervisor; the steps are detailed for VirtualBox.

Using the same computer that you used in Capstone Project 1-1, complete the following steps:

1. Go to **ubuntu.com** and download the latest LTS (long-term support) version of the Ubuntu Desktop OS to your hard drive. This is a free download, so you can decline to make any donations. The file that downloads is an ISO file.
2. Open the Oracle VM VirtualBox Manager. Following the directions in the Chapter 1 Capstone Project, create a new VM with an informative name. Consider the following tips:
   a. For ISO Image, select the ISO file that contains the Ubuntu Desktop image to a virtual DVD in your VM.
   b. If necessary, choose the **Linux** type and the **Ubuntu (64-bit)** version.
3. As the VM starts, it begins to install Ubuntu Desktop. Record your user credentials for your Ubuntu VM in a secure note in your Bitwarden vault. Once the OS is fully installed, sign in. You don't need to connect any online accounts or enable Ubuntu Pro. Other options are up to you.
4. To verify you have an Internet connection, open the Mozilla Firefox browser and surf the web. **Take a screenshot** of your desktop showing your hypervisor, your running VM, and the VM's successful connection with the Internet; submit this visual with your responses to this project's questions.
5. Good network technicians must know how to use many operating systems. Poke around in the Ubuntu Desktop interface and get familiar with it. You can also search the web for tutorials and YouTube videos on how to use Ubuntu Desktop. What are two ways to open the Settings window in Ubuntu Desktop?
6. When you're ready to shut down your VM, click the power icon in the upper-right corner of the Ubuntu Desktop screen, click **Power Off/Log Out**, click **Power Off ...**, and then click **Power Off**.
7. Before you walk away from this project, take a moment to add the new information to your VMclients page in your wiki. Go to the **Virtualization:VMclients** page, click **Edit** at the bottom of the page, and add the new VM to your list. Include the chapter number, hypervisor used, VM computer name, and VM operating system. Also note any additional information that you might find helpful when you return to this VM in the future. Do *not* include sensitive information, such as username and password—this information should be recorded only in a safe place, such as your Bitwarden vault. When you're finished, click **Save**.
8. **Take a screenshot** of the edited wiki page; submit this visual with your responses to this project's questions.

## Capstone Project 2-2: Install and Use Packet Tracer

**Estimated time:** 45 minutes

**Objective 3.1:** Explain the purpose of organizational processes and procedures.

**Resources:**

- A computer or user account with application installation rights
- Internet access

**Description:**

If you plan to pursue networking or cybersecurity as your area of specialty in IT, you might consider earning a few Cisco networking certifications after you complete your CompTIA Network+ certification.

Cisco's Skills for All website provides many useful tools for advancing your networking education. One of those tools is a network simulator called Packet Tracer.

In this Capstone Project, you download and install Packet Tracer and take a tour of the simulator interface. This version of Packet Tracer is free to the public, and your school does not have to be a member of Cisco's Networking Academy for you to download and use it. In later projects, you'll return to Packet Tracer to build networks and even learn some basic Cisco IOS (Internetworking Operating Systems) commands. Cisco IOS is the operating system used on Cisco networking devices, such as routers and switches (with minor variations in the specific IOS for each different type of device). Many other manufacturers of networking devices use the same or similar commands, and those that use different commands typically use very similar functions, even if they call it something a little different.

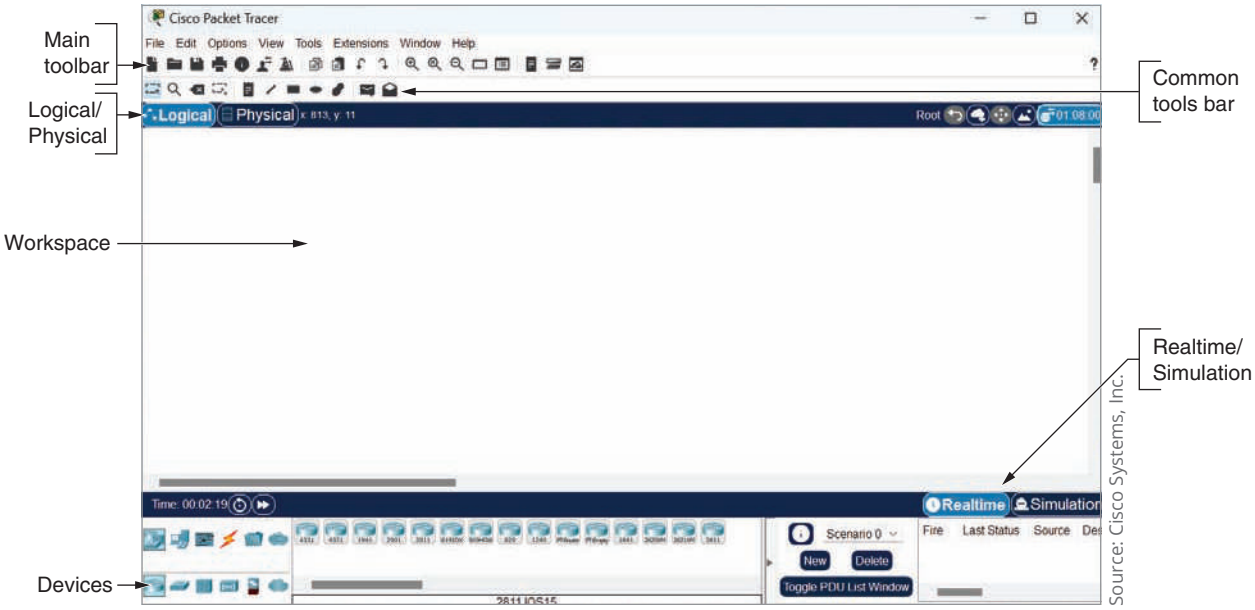
To get the Packet Tracer download, you must first sign up for the free Introduction to Packet Tracer online course on Cisco's Skills for All website. Complete the following steps to create your account:

1. Go to **skillsforall.com/learningcollections/cisco-packet-tracer**. If the course is not listed on this page, do a search for **packet tracer site:skillsforall.com** and follow links to "Download Packet Tracer" or "Introduction to Packet Tracer" to find the current Packet Tracer introduction course. Enroll in the free Getting Started with Cisco Packet Tracer course. If you don't have an account with Skills for All or with Cisco Networking Academy, sign up for a new account or use a Google account.
2. Save your account information in your Bitwarden vault. You will need this information again.

Now you're ready to download and install Packet Tracer. If you need help with the download and installation process, check the Packet Tracer course resources for additional guidance. Complete the following steps:

3. Inside the course, find the link to download Packet Tracer. Choose the correct version for your computer.
4. After the download is complete, install Packet Tracer. When the installation is complete, run **Cisco Packet Tracer**. When Packet Tracer asks if you would like to run multi-user, click **No**. When asked if you want to allow public and private networks to access this app, click **Cancel**.
5. When Packet Tracer opens, sign in with your Skills for All account that you just created. If you see a Windows Security Alert, allow access through your firewall. Cisco Packet Tracer opens (Figure 2-48).

**Figure 2-48** Explore the Packet Tracer window




Source: Cisco Systems, Inc.

The Introduction to Packet Tracer course presents an excellent introduction to Packet Tracer and provides lab activities. Packet Tracer activities are interactive labs in which you download a start file, make the changes instructed in the lab, and then grade the activity in Packet Tracer. Complete the following steps to access your course:

6. Return to your Introduction to Packet Tracer course.
7. Watch the Packet Tracer overview video and the two videos for the Packet Tracer interface. The other lessons provide excellent information on Packet Tracer but are not required for this project. Respond to the following questions along the way:
   a. Which mode illustrates how data will flow through the network?
   b. Which box lets you create packets to travel across the network?
   c. Which Packet Tracer feature do you think will be most helpful for you in learning how to manage a network? Why do you think this?
8. Back in your Packet Tracer window, deploy at least three physical devices into your workspace. You don't need to configure or connect them. **Take a screenshot** of your deployment; submit this visual with your responses to this project's questions.
9. In your wiki, add a new page titled **Applications:PacketTracer**. Remember *not* to include a space after the colon so your PacketTracer page will be collected in the Applications category. Indicate the chapter and project number for this installation, the computer you used for this project, a brief description of what you learned, and any other information you might find helpful when using Packet Tracer later. You'll return to Packet Tracer many times throughout this course.

**Note to Instructors:** For later chapters, a Packet Tracer solution file is provided for each Packet Tracer project through the Cengage Instructors site. Some Packet Tracer projects build on earlier Packet Tracer networks. If needed for one or more students, you can provide a previous project's solution file as a start file for one of these progression projects.

# Solutions to Self-Check Questions

## Section 2-1: Components of Structured Cabling

1. At what point does the ISP hand over responsibility to the customer?
   **Answer**: c. Demarc
   **Explanation**: For most situations, the device that marks where a telecommunications service provider's network ends and the organization's network begins is the **demarc**, or demarcation point. The MDF (main distribution frame) and IDF (intermediate distribution frame) are environmentally controlled spaces holding network equipment. While the demarc might be located inside the MDF, the demarc itself is the cutoff point of responsibility. A patch panel is a panel of data receptors which can be mounted to a wall or a rack and which provides a central termination point when many patch cables converge in a single location.

2. According to structured cabling standards, there must be at least one _____ on each floor.
   **Answer**: a. IDF
   **Explanation**: The ANSI/TIA standard specifies at least one **IDF (intermediate distribution frame)** per floor, although large organizations may have several data rooms or closets per floor to better manage the data feed from the main data facilities. There is only one MDF (main distribution frame) per campus and one demarc per ISP (Internet service provider). There can be none, one, or several work areas per floor, and the number of work areas is not defined by structured cabling standards.

3. Which of the following causes physical layer failures?
**Answer**: d. EMI
**Explanation**: Install cable at least 3 feet away from fluorescent lights or other sources of **EMI (electromagnetic interference)** to reduce the possibility of noise (interference) that can affect your network's signals at the physical layer. A SCADA (supervisory control and data acquisition) system acquires real-time data from a physical system and manages the system or presents the data to humans to monitor and manage the system. Regular twisted-pair cables are coated with PVC (polyvinyl chloride), which is made from a cheap plastic that is toxic when burned. STP (shielded twisted-pair) cable is a copper-based cable containing twisted-pair wires with metallic shielding, such as foil around each wire pair or surrounding all four wire pairs.

## Section 2-2: Network Documentation

4. Which network diagram shows a logical topology?
**Answer**: b. Network map
**Explanation**: Network mapping is the process of discovering and identifying the devices on a network. A **network map** shows logical connections and addressing information. Diagrams of physical layouts include floor plans and rack diagrams. A baseline configuration is a change management concept that defines a stable state to be achieved and maintained before attempting future changes.

5. What documentation will indicate how many copies you can install of a particular application?
**Answer**: c. License
**Explanation**: **Licensing restrictions** might define who is allowed to use an application and for how long, how many users are allowed to install or access it, whether the application can be made available over a company's network or the Internet, and how many backup copies of the application may be stored. A baseline configuration defines a stable state to be achieved and maintained in a system before attempting future changes. SOPs (standard operating procedures) help ensure consistency as different employees perform the same complex tasks. An SLA (service level agreement) is a legally binding contract or part of a contract that defines, in plain language and in measurable terms, the aspects of a service provided to a customer.

6. When an Internet connection goes down, what business document defines a customer's options for paying a reduced rate?
**Answer**: d. SLA
**Explanation**: An **SLA (service-level agreement)** specifies options for compensation if outages exceed defined thresholds. An MOU (memorandum of understanding) documents the intentions of two or more parties to enter into a binding agreement. An SOW (statement of work) documents in detail the work that must be completed for a particular project. An RFP (request for proposal) is a request to vendors to submit a proposal for a product or service your company wants to purchase.

## Section 2-3: Change Management

7. Which is longer for a reliable device?
**Answer**: a. MTBF
**Explanation**: The **MTBF (mean time between failures)** is the average amount of time that will pass for devices before the next failure is expected to occur. This should be significantly longer than the MTTR (mean time to repair), which is the average amount of time required to repair a device once it fails. A SPoF (single point of failure) is a single device, connection, service, or any other resource on which other resources depend and which has no backup. SSOT (single source of truth) refers to information gathered into a central repository.

8. Which change is designed to close a security vulnerability without affecting other components of an application?
**Answer**: a. Patch
**Explanation**: A software **patch** is a correction, improvement, or enhancement to software. It corrects a bug, closes a vulnerability, or adds minor enhancements to only part of the software, leaving most of the code untouched. A software rollback is the process of reverting to a previous version of software after attempting to patch or upgrade it. A software upgrade is a major change to a software package that enhances the functionality and features of the software, while also correcting bugs and vulnerabilities. New software is installed on a device.

9. Which change management principle must you apply to all changes regardless of scope or impact?
**Answer**: d. Document the change.
**Explanation**: Some change management principles apply to any type of change, including the requirement to **document each change** throughout the process. More extensive or expensive changes typically must progress through a formal change management process, which might include submitting a change request document and acquiring review board approval. Patches or upgrades should always be thoroughly tested and planned, although updates that close a critical security vulnerability should be prioritized and applied as soon as reasonably possible.